Introduction & Problem Statement w/Ani, Robby, Diana, and Julian

The goal of this project is to examine how hospital location and payer type influence price variation for Radiation Oncology procedures across Florida hospitals. Hospital price transparency regulations, introduced by CMS, require hospitals to publish machine-readable files containing payer-specific negotiated rates and most valuable information about the true economic landscape of healthcare pricing. **Radiation Oncology** was selected because it is one of the most expensive components of cancer treatment, and patients commonly face varying, seemingly unpredictable costs. Negotiated rates for the same CPT code can vary dramatically across both different hospitals and payer groups. Understanding the sources of variation is crucial for identifying inequities, evaluating market power, and informing both policy and patient decision-making.

Research Question

"Is there a meaningful relationship between hospital location and the negotiated price variation of Radiation Oncology CPT procedures across Medicare, Medicaid, Private/Commercial, and Self-Pay payer groups in Florida?"

This research question is important because it enables us to explore the following:

- Whether certain cities exhibit consistently higher or more volatile pricing
- Whether payer groups face systematically different negotiated rates
- Whether specific treatment categories within Radiation Oncology show more pricing dispersion
- Whether market dynamics differ by geography or payer mix

Data Description

Our dataset originates from the machine-readable hospital pricing files published by Florida hospitals. These files may be in wide or tall format and include information such as:

- CPT/HCPCS code
- Payer name and Plan name
- Negotiated rate (payer-specific) and Estimated Amount
- Gross charge
- Cash price/self-pay charge
- Hospital name and address

We use only the CPT codes relevant to Radiation Oncology within the range of 77261–77799. These codes were mapped into higher-level treatment categories (e.g., Simulation, IMRT, SBRT, Treatment Delivery, Brachytherapy). Our extraction revealed that ~45 relevant CPT codes are present across ~45 unique hospitals in Florida. A city field was created by parsing the hospital address. This allows for geographic and market-level analysis and is essential for our research question.

Research Plan & Methodology

Our methodology consists of three major components: ETL and database loading, data preprocessing, and statistical + geospatial analysis.

ETL & Database Loading

We developed a multi-step pipeline that loads raw hospital pricing files into PostgreSQL performing a Wide → Tall Transformation. Using 2_wide_to_tall_to_pg.py, wide-format machine-readable files are transformed into a tall structure. It reads and normalizes raw columns, converts payer-specific columns into row-level payer entries, creates unique identifiers, and then handles chunked loading into PostgreSQL

Numeric Field Extraction

With load_numeric_fields.py, we extract Negotiated rates, Gross charges, Cash prices, Percentage-of-charge calculations. All fields are standardized and merged into a structured database table.

Data Preprocessing (using 3_refined_data.py)

Cleaning, Creating, and Filtering Columns

After extracting from the database, the CPT codes needed to be consolidated into one column and filtered for not only the Radiation Oncology range (77261–77799) but also for codes that appear in 50% or more of unique hospitals. Also creating a new column ["City"] for Florida cities.

Payer & Plan Standardization

Due to inconsistency in payer fields, we create a unified payer_text field combining payer_name, plan_name, and additional notes.

Payer Group Classification & CPT Mapping

We apply rule-based classification on Medicare, Medicaid, Private / Commercial, Self-Pay/Other, & Unknown. This enables consistent comparison across hospitals. We map CPT codes into Treatment Categories using a provided mapping file.

Missingness Analysis & Imputation

Missing negotiated prices were not random: they varied by CPT category and payer group. Dropping missing values would cause biased estimates; global imputation would distort true pricing.

We therefore used Stratified Median Imputation, following a 3-step hierarchy:

- Median within (Treatment Category × Payer Group)
- Median within Treatment Category
- Global median fallback

This preserves the real cost structure of oncology pricing.

Outlier Analysis

We tested outlier removal and capping but determined that modifying extreme values would distort valid price variation—an essential outcome of our research question. Instead, we detected outliers using the IQR method and flagged them, keeping them for analysis.

Final Dataset

We generate 4_cleaned_prices.csv, which becomes our analysis-ready dataset.

Statistical & Geospatial Analysis Plan

Descriptive Statistics

We computed summary statistics at multiple granularities; here are a few examples concerning the descriptive statistics of Negotiated Price:

PRICE SUMMARY BY CPT CODE

CPT Category	N	Mean	Median	Std Dev	CV%
Clinical Brachytherapy Radiation Treatment	17,766	$1,745.12	$941.18	$11,245.38	644.4%
Clinical Treatment Planning (External & Internal)	9,453	$1,716.57	$1,238.00	$8,676.87	505.5%
Medical Radiation Physics, Dosimetry, Treatment	26,025	$1,083.87	$679.00	$7,918.48	730.6%
Neutron Beam Treatment Delivery	1,205	$13,287.61	$9,312.00	$71,372.40	537.1%
Radiation Treatment Delivery	10,583	$601.86	$420.00	$1,988.40	330.4%
Radiation Treatment Management	1,973	$1,625.37	$1,537.00	$5,538.70	340.8%
Stereotactic Radiation Treatment Delivery (SBRT)	11,430	$5,783.73	$1,808.00	$32,138.63	555.7%

PRICE SUMMARY BY PAYER TYPE

Payer Type	N	Mean	Median	Std Dev	CV%
Medicaid	9,460	$831.92	$125.53	$1,976.06	237.5%
Medicare	19,726	$1,105.77	$648.00	$1,657.31	149.9%
Private / Commercial	48,963	$2,797.33	$992.00	$21,594.34	772.0%
Self-Pay / Other	286	$1,699.90	$572.00	$5,796.97	341.0%

PRICE SUMMARY BY LOCATION

City	N Hospitals	N Records	Mean Price	Median Price	CV%
Tarpon Springs	3,850	3,850	$6,588.57	$936.00	1133.1%
Daytona Beach	1,000	1,000	$3,594.97	$828.50	336.2%
Dunedin	2,312	2,312	$1,315.35	$363.95	282.2%
Petersburg	1,888	1,888	$1,475.49	$476.20	274.9%
Safety Harbor	1,888	1,888	$1,475.49	$476.20	274.9%

City	N Hospitals	N Records	Mean Price	Median Price	CV%
Clearwater	**1,888**	**1,888**	**$1,475.49**	**$476.20**	**274.9%**
New Port Richey	**3,776**	**3,776**	**$1,475.49**	**$476.20**	**274.8%**
Tallahassee	**686**	**686**	**$1,773.63**	**$758.10**	**265.2%**
Port Charlotte	**983**	**983**	**$2,146.63**	**$758.10**	**239.2%**
Ft. Walton Beach	**589**	**589**	**$2,533.34**	**$1,238.00**	**227.3%**

Geospatial Mapping

Using 6_geospatial_visual.py, we create 11 interactive folium maps:

- Overall price landscape
- Price variation hotspots
- Top CPTs (6 maps)
- Payer disparity map
- Data coverage / market size map

Preliminary Findings (High-Level) & Next Steps

Based on early analysis:

- Radiation Oncology prices vary significantly across Florida cities.
- Private insurers consistently pay more than Medicare and Medicaid.
- Advanced modalities (IMRT, SBRT) show the highest variation.
- Cities like Orlando and Jacksonville appear to be high-variation markets.
- Keeping outliers was justified, as they represent real market behavior.

Next Steps

- Complete full descriptive and geospatial analysis
- Explore regression approaches to quantify location and payer effects
- Develop final insights & recommendations

Conclusion

This project provides a comprehensive view of price variation in Radiation Oncology across Florida. By integrating transparent ETL, rigorous preprocessing, and robust analytical techniques, we aim to uncover meaningful geographic and payer-driven pricing patterns. The results will contribute to understanding fairness, variation drivers, and transparency in cancer treatment pricing.